Lesson Planning

Complete Lesson Plan Book

Class: FSc Pre-Medical · Class 12

Subject: Biology

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Human Nervous System: Neurons and Reflex Action

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Unit 1: Human Nervous System: Neurons and Reflex Action

Learning Objectives

• Students will describe the structure of a neuron and the function of each part.

• Students will trace the path of a nerve impulse through a reflex arc.

• Students will explain why reflex actions are faster than voluntary actions.

Key Concepts

• A neuron is the basic unit of the nervous system; it has dendrites that receive signals, a cell body, and an axon that carries the impulse away.

• A nerve impulse travels as an electrochemical signal along the axon and jumps from one neuron to the next across a synapse.

• A reflex arc runs from a receptor through a sensory neuron, the spinal cord and a motor neuron to an effector, without waiting for the brain.

• The knee-jerk and the hand pulled away from a hot object are reflex actions that protect the body.

• Reflex actions are fast because the pathway is short and the spinal cord handles the response directly.

Teaching Methodology (Step by Step)

1. Ask what happens when a hand touches something hot, and why the hand moves before we think about it.

2. Draw a large neuron on the board and label dendrites, cell body, axon and synapse.

3. Trace a reflex arc step by step, from the receptor in the skin to the muscle that pulls the hand away.

4. Contrast this short pathway with a voluntary action that must travel to the brain and back.

5. Test understanding with the knee-jerk example and ask students to name each part of its arc.

Classroom Activities

• Diagram labelling: students label a neuron diagram and a reflex arc diagram in pairs.

• Role play: five students act as receptor, sensory neuron, spinal cord, motor neuron and muscle in sequence.

• Case discussion: students explain why a person with a damaged sensory neuron might not feel a hot surface.

Assessment Questions

1. Label the parts of a neuron and state the function of the axon.

2. Put these in order: effector, receptor, motor neuron, sensory neuron, spinal cord.

3. Why is a reflex action faster than picking up a pen on purpose?

Homework

• Draw and label a reflex arc for touching a hot plate.

• Write three examples of reflex actions from daily life and name the effector in each.

Time Allocation

45 minutes